

3.STRING MATCHING:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char text[100], pattern[100];
```

```
    int i, j;
```

```
    printf("Enter the text: ");
```

```
    scanf("%s", text);
```

```
    printf("Enter the pattern: ");
```

```
    scanf("%s", pattern);
```

```
    int n = strlen(text);
```

```
    int m = strlen(pattern);
```

```
    for (i = 0; i <= n - m; i++) {
```

```
        for (j = 0; j < m; j++) {
```

```
            if (text[i + j] != pattern[j])
```

```
                break;
```

```
        }
```

```
        if (j == m)
```

```
            printf("Pattern found at index %d\n", i);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

main.c	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char text[100], pattern[100]; 6 int i, j; 7 8 printf("Enter the text: "); 9 scanf("%s", text); 10 11 printf("Enter the pattern: "); 12 scanf("%s", pattern); 13 14 int n = strlen(text); 15 int m = strlen(pattern); 16 17 for (i = 0; i <= n - m; i++) { 18 for (j = 0; j < m; j++) { 19 if (text[i + j] != pattern[j]) 20 break; 21 } 22 }</pre>	<pre>Enter the text: ABABDABACDABABCABAB Enter the pattern: ABABCABAB Pattern found at index 10 === Code Execution Successful ===</pre>